

```
In [6]: import numpy as np
import pandas as pd
import warnings; warnings.filterwarnings("ignore")
```

```
In [7]: from sklearn.datasets import load_iris

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)

print("The shape of data is",df.shape)
df.head()
```

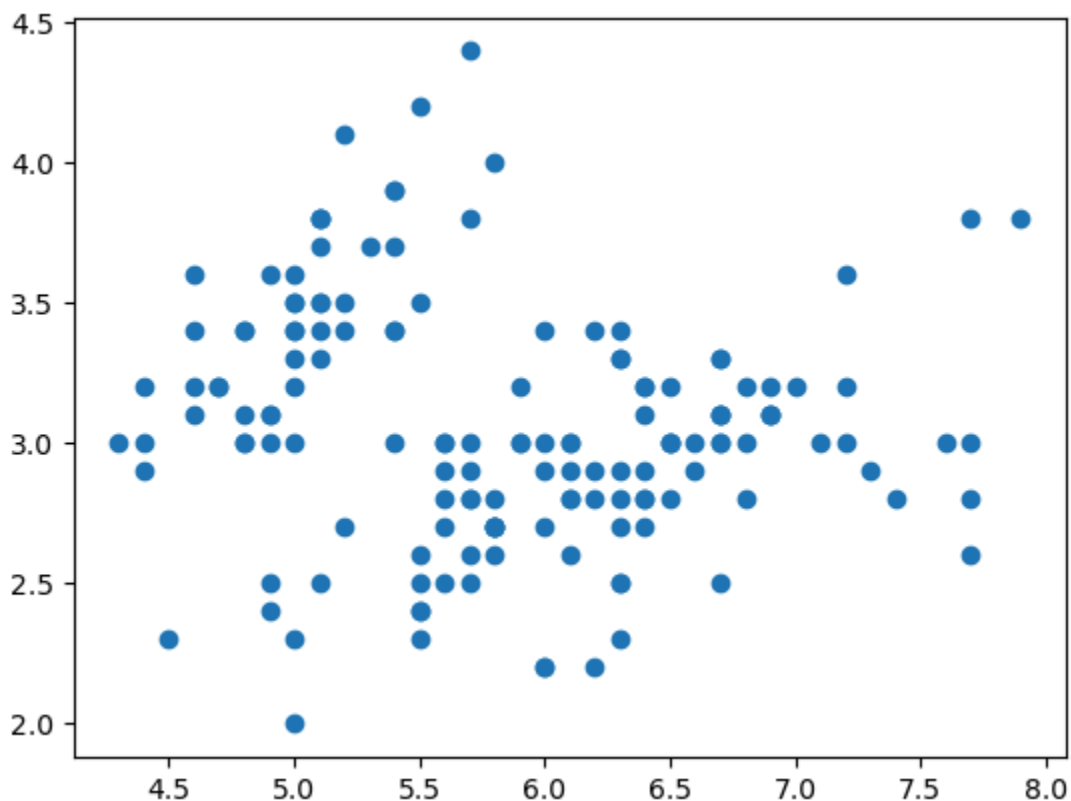
The shape of data is (150, 4)

```
Out[7]:
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

```
In [8]: import matplotlib.pyplot as plt
plt.scatter(df.iloc[:,0],df.iloc[:,1])
```

Out[8]: <matplotlib.collections.PathCollection at 0x2b528095590>



```
In [9]: from sklearn.cluster import KMeans
```

```
In [10]: wcss = []

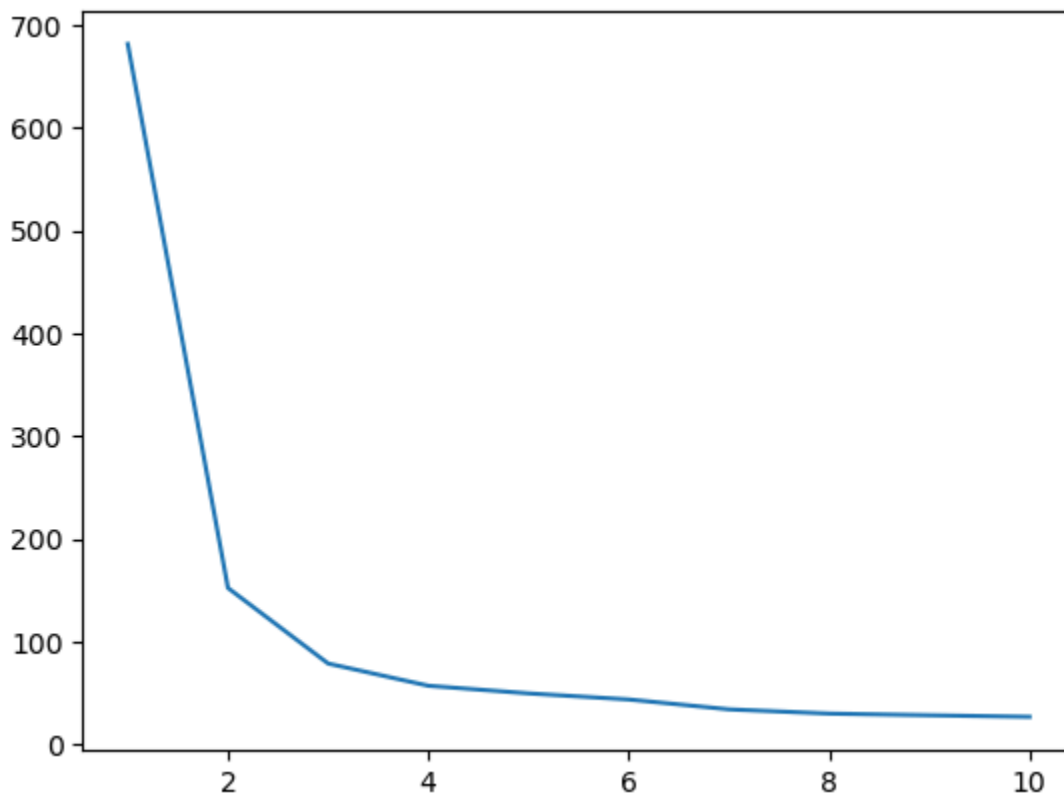
for i in range(1,11):
    km = KMeans(n_clusters=i)
    km.fit_predict(df)
    wcss.append(km.inertia_)
```

```
In [11]: wcss
```

```
Out[11]: [681.3706,
          152.34795176035792,
          78.85144142614601,
          57.25600931571816,
          49.827740558292284,
          43.91240257473925,
          34.29971212121212,
          30.185072738664847,
          28.560733484781473,
          27.04915282137883]
```

```
In [12]: plt.plot(range(1,11),wcss)
```

```
Out[12]: [<matplotlib.lines.Line2D at 0x2b5288b4e10>]
```



```
In [13]: X = df.iloc[:, :].values
km = KMeans(n_clusters=3)
```

```
y_means = km.fit_predict(X)
```

```
In [14]: y_means
```

```
Out[14]: array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
               0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
               0, 0, 0, 0, 0, 0, 1, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,  
               2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,  
               2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 1, 1, 1, 1, 2, 1, 1, 1,  
               1, 1, 1, 2, 2, 1, 1, 1, 1, 2, 1, 2, 1, 2, 1, 1, 2, 2, 1, 1, 1, 1,  
               1, 2, 1, 1, 1, 1, 2, 1, 1, 1, 2, 1, 1, 1, 2, 1, 1, 2], dtype=int32)
```

```
In [15]: X[y_means == 2,1]
```

```
Out[15]: array([3.2, 2.3, 2.8, 2.8, 3.3, 2.4, 2.9, 2.7, 2. , 3. , 2.2, 2.9, 2.9,
                3.1, 3. , 2.7, 2.2, 2.5, 3.2, 2.8, 2.5, 2.8, 2.9, 3. , 2.8, 2.9,
                2.6, 2.4, 2.4, 2.7, 2.7, 3. , 3.4, 3.1, 2.3, 3. , 2.5, 2.6, 3. ,
                2.6, 2.3, 2.7, 3. , 2.9, 2.9, 2.5, 2.8, 2.7, 2.5, 2.5, 2.8, 2.2,
                2.8, 2.7, 2.8, 3. , 2.8, 3. , 2.7, 2.5, 3. ])
```

```
In [16]: plt.scatter(X[y_means == 0,0],X[y_means == 0,1],color='blue')
plt.scatter(X[y_means == 1,0],X[y_means == 1,1],color='red')
plt.scatter(X[y_means == 2,0],X[y_means == 2,1],color='green')
```

```
Out[16]: <matplotlib.collections.PathCollection at 0x2b528927890>
```

